

Scenario

Two different ONOS instances are deployed in a distributed cluster for high availability and scalability. They need to share network state and coordinate control plane decisions.

Question: Which ONOS interface facilitates this communication and synchronization between the ONOS instances?

Answer:

Communication and state synchronization between ONOS instances in a distributed cluster is facilitated by the **East-West (East/West) Interface**, utilizing the internal **ONOS Core Distributed Store Messaging Service (Atomix / Raft Consensus Protocol)**.

Explanation:

- **East-West Interface:** Connects multiple SDN controllers or distributed instances within a cluster to synchronize cluster topology, device mastership, dynamic flow states, and event notifications.
- **Atomix & Raft Consensus:** ONOS uses an underlying distributed coordination store (Atomix) operating over the East-West interface to achieve strong consistency, anti-entropy state synchronization, and continuous high availability across all controller nodes.